AI Assisted Coding

Lab Assignment - 16.1

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Lab 16 – Database Design and Queries: Schema Design and SQL Generation

Lab Objectives:

- Understand schema design principles for relational databases.
- Use AI-assisted tools to generate SQL schema definitions.
- Write and refine SQL queries (CRUD + aggregate operations).
- Explore how AI can optimize queries and ensure normalization.

Task 1 – Student Information System Schema

Task: Design a database schema for a Student Information System and generate queries using AI.

Instructions:

- Tables: Students, Courses, Enrollments.
- Define primary keys, foreign keys, and relationships.
- Generate queries:
    - o Insert a new student record.
    - o Fetch all courses enrolled by a student.
    - o Count number of students in each course.

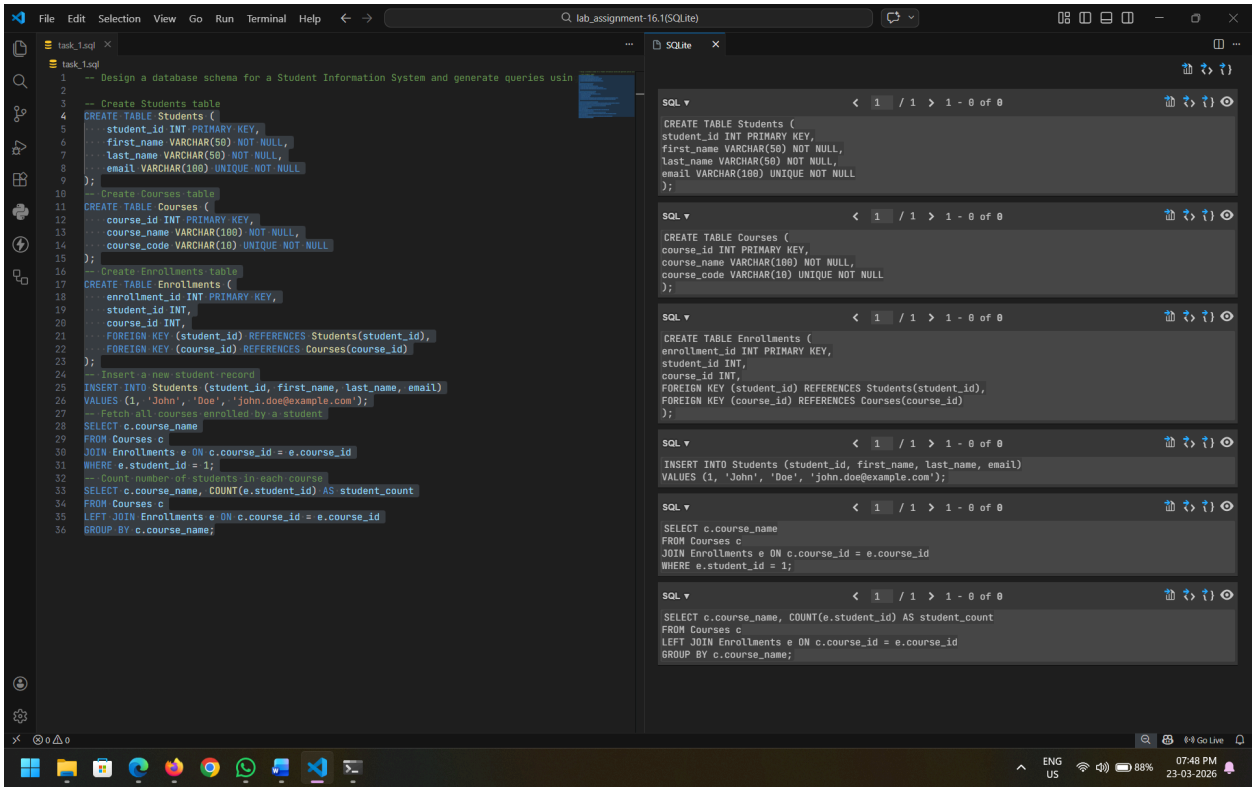
Expected Output:

• SQL CREATE TABLE statements and queries for student– course relationships.

Firstly we have to create a database file :

```
C:\SQLite>sqlite3 task_1.db
SQLite version 3.51.3 2026-03-13 10:38:09
Enter ".help" for usage hints.
sqlite> .database
main: C:\SQLite\task_1.db r/w
sqlite>
```

Then we have to run the .sql file, So that we get the output in this .db file:



Task 2 – Hospital Management Database

Task: Create schema and queries for a Hospital Management System.

Instructions:

- Tables: Doctors, Patients, Appointments.
- Use AI to define constraints (unique IDs, valid dates).
- Generate queries:
    - o List all appointments for a specific doctor.
    - o Retrieve patient history by patient ID.
    - o Count total patients treated by each doctor.

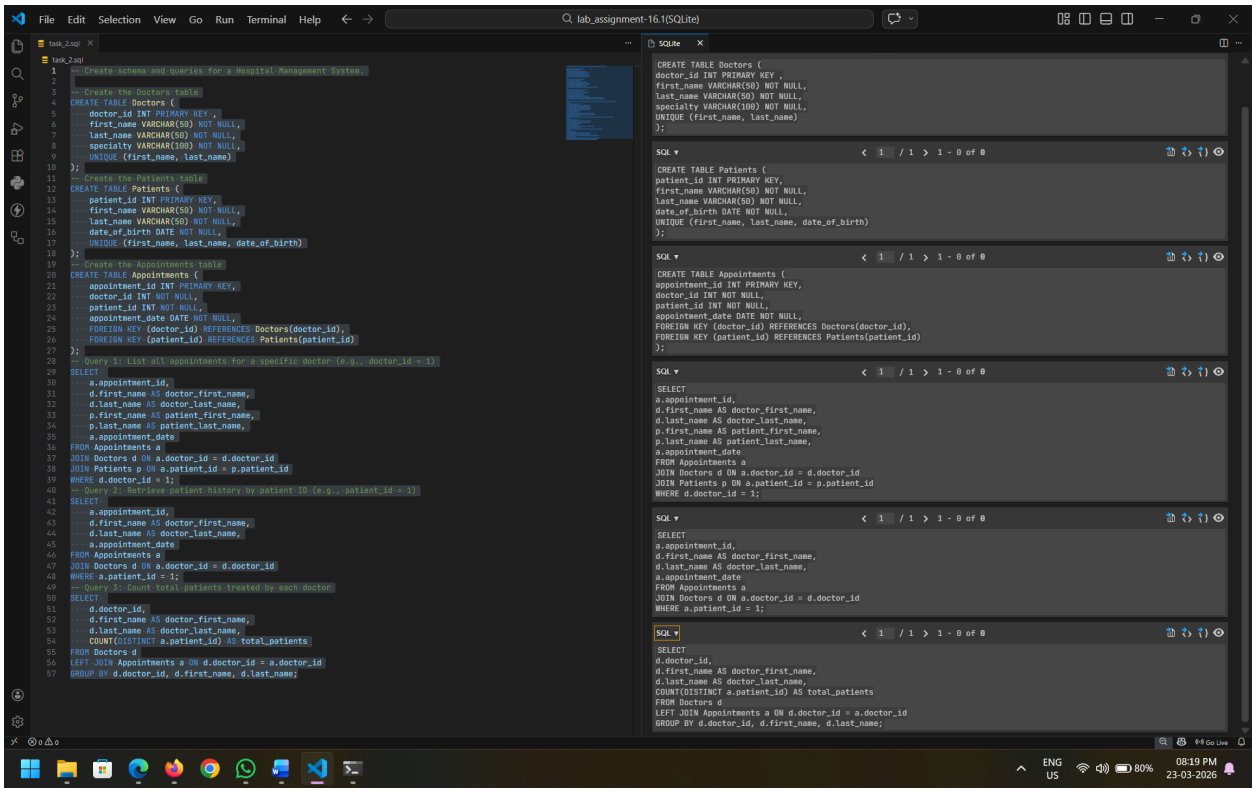
Expected Output:

- Normalized schema and SQL queries with joins.

Firstly we have to create a database file :

```
C:\SQLite>sqlite3 task_2.db
SQLite version 3.51.3 2026-03-13 10:38:09
Enter ".help" for usage hints.
sqlite> .database
main: C:\SQLite\task_2.db r/w
sqlite>
```

Then we have to run the .sql file, So that we get the output in this .db file:



Task 3 – Library Database

Task: Design schema for a Library Management System.

Instructions:

- Tables: Books, Members, Loans.
- Use AI to suggest indexing strategy for faster queries.
- Generate queries:

o Retrieve all books currently issued.
o Find overdue books (loan date > 30 days).
o Count number of books loaned by each member.

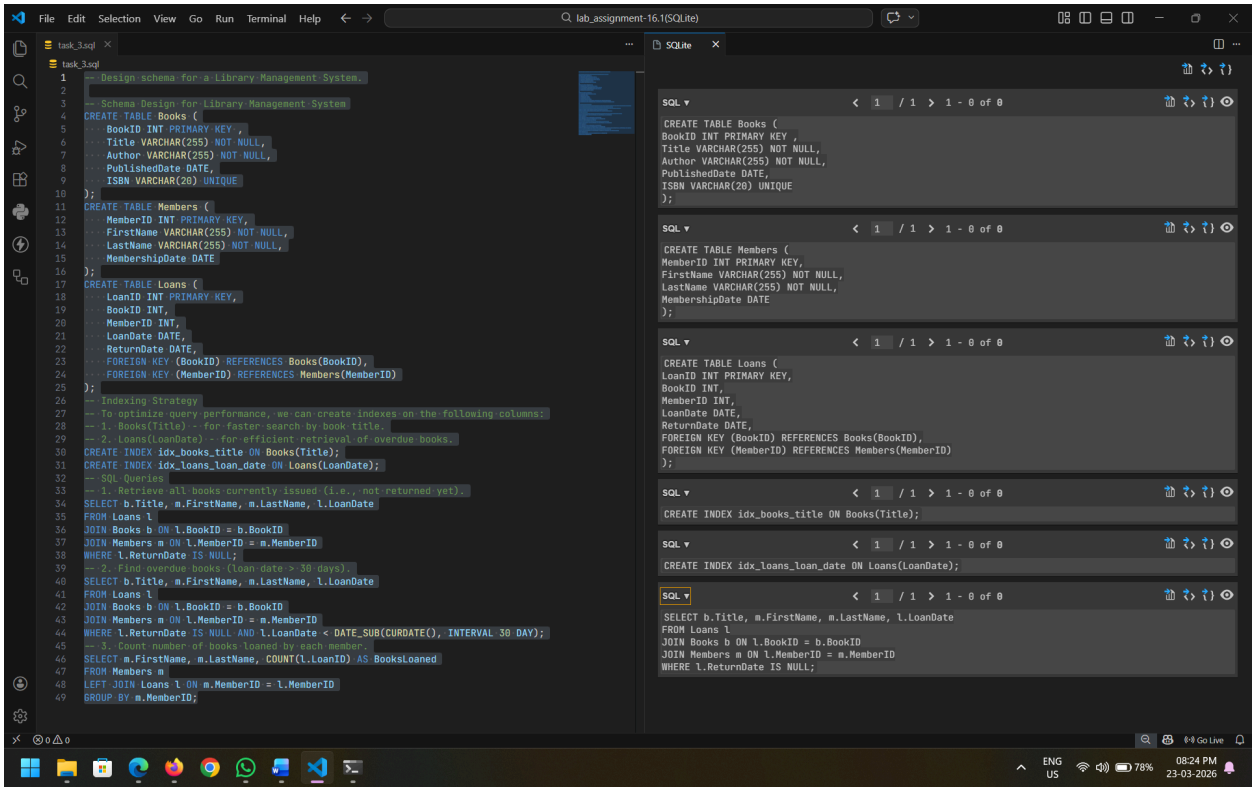
Expected Output:

• Schema + SQL queries demonstrating joins and conditions.

Firstly we have to create a database file :

```
C:\SQLite>sqlite3 task_3.db
SQLite version 3.51.3 2026-03-13 10:38:09
Enter ".help" for usage hints.
sqlite> .database
main: C:\SQLite\task_3.db r/w
sqlite>
```

Then we have to run the .sql file, So that we get the output in this .db file:



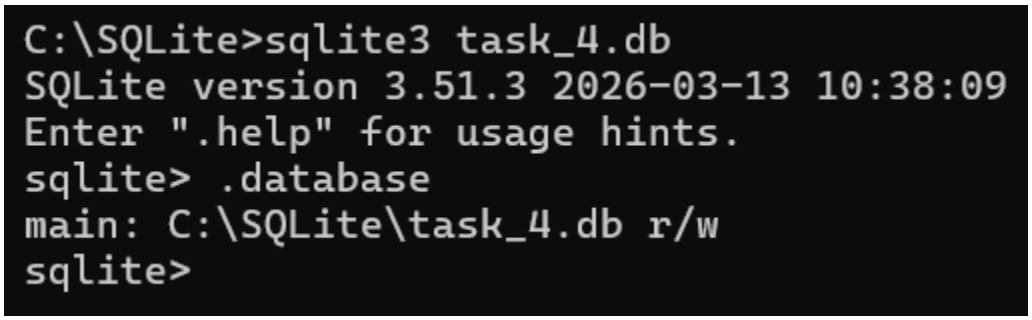
Task 4 – Real-Time Application: Online Shopping Database
Scenario: Design a database for an E-commerce platform.
Requirements:

- Tables: Users, Products, Orders, OrderDetails.
- Generate queries to:
  - Retrieve all orders by a user.
  - Find the most purchased product.
  - Calculate total revenue in a given month.
- AI should also suggest normalization improvements and query optimization.

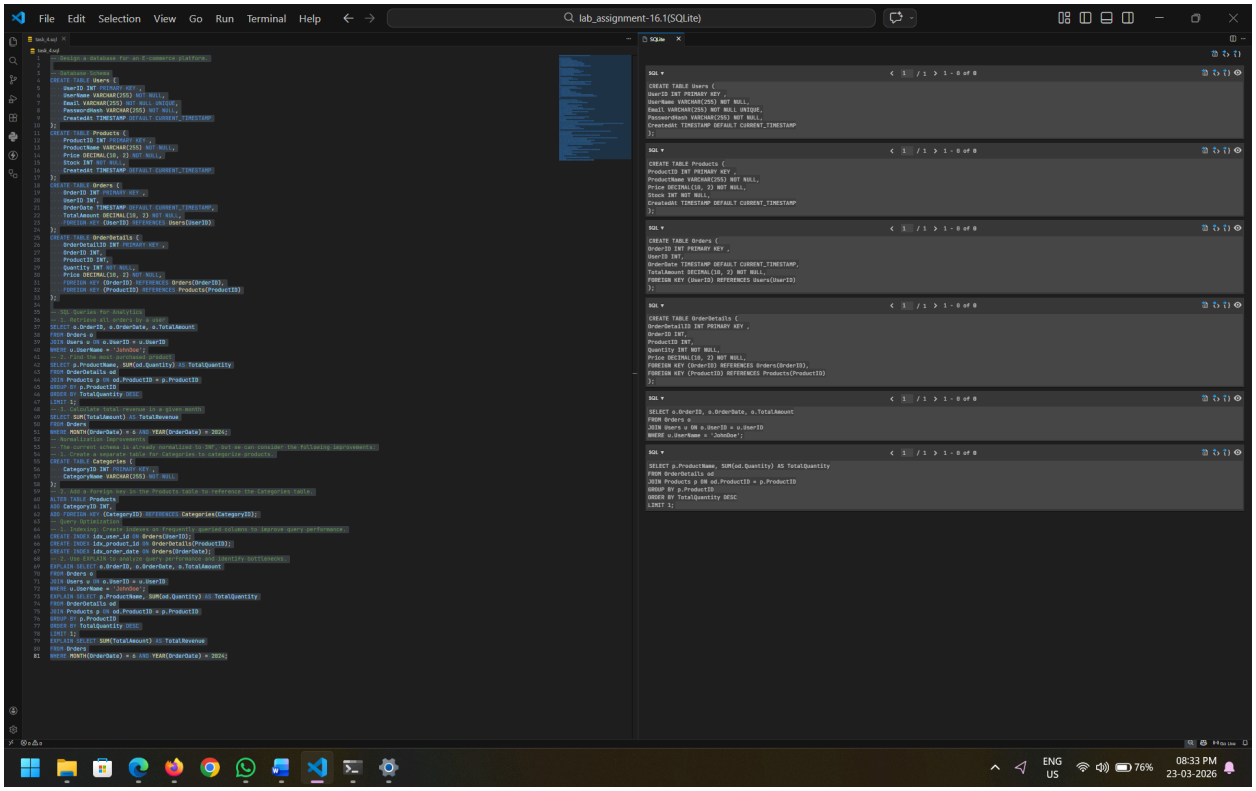
Expected Output:

- Complete schema with relationships + SQL queries for analytics.

Firstly we have to create a database file :



Then we have to run the .sql file, So that we get the output in this .db file:



Task 5 – University Examination Database

Design a database schema for a University Examination System and generate SQL queries using AI.

Instructions:

- Tables: Students, Subjects, Exams, Results.
- Define primary keys, foreign keys, and referential integrity constraints.
- Generate queries:
    - o Insert examination results for a student.
    - o Retrieve subject-wise marks for a student.
    - o Calculate average marks for each subject.

Expected Output:

- Normalized schema and SQL queries involving aggregation and joins.

Firstly we have to create a database file :

```
C:\SQLite>sqlite3 task_5.db
SQLite version 3.51.3 2026-03-13 10:38:09
Enter ".help" for usage hints.
sqlite> .database
main: C:\SQLite\task_5.db r/w
sqlite>
```

Then we have to run the .sql file, So that we get the output in this .db file: